Question	Answer	Marks	Partial Marks
1(a)(i)	$5.101[00\ldots] \times 10^8$ final answer	**1**	
1(a)(ii)	361 150 800 oe	**2**	**FT** *their* **(a)(i)** **M1** for $\frac{70.8}{100} \times 510\,100\,000$ or for $\frac{70.8}{100} \times$ *their* **a(i)**
1(b)(i)	6070 oe	**1**	
1(b)(ii)	32 000 oe	**2**	**B1** for figs 32
1(b)(iii)	6.68 or 6.677 …	**2**	**M1** for $\frac{6.41\times10^5}{9.6[0]\times10^6}$ [× 100] oe
1(b)(iv)	1250 or 1248 to 1249 oe	**2**	**B1** for figs 125 or figs1248 to figs 1249
1(c)(i)	25.1 or 25.08…	**2**	**M1** for $\frac{7.53\left[\times10^9\right]-6.02\left[\times10^9\right]}{6.02\left[\times10^9\right]}$ oe or $\frac{7.53\left[\times10^9\right]}{6.02\left[\times10^9\right]}\times100$
1(c)(ii)	1.33 or 1.325…	**3**	**M2** for $\sqrt[17]{\frac{7.53[\times10^9]}{6.02[\times10^9]}}$ or $\sqrt[17]{1+\frac{\textit{their}\ \textbf{(c)(i)}}{100}}$ or **M1** for $6.02[\times10^9] \times p^{17} = 7.53[\times10^9]$ or $p^{17} = 1+\frac{\textit{their}\ \textbf{(c)(i)}}{100}$
2(a)(i)	Triangle at (–3, 2) (–3, 3) (–5, 2)	**2**	**B1** for correct rotation about incorrect point or for rotation 90 clockwise around (0, 0)
2(a)(ii)	Triangle at (5, –2) (6, –2) (5, 0)	**2**	**B1** for translation by $\begin{pmatrix}3\\k\end{pmatrix}$ or $\begin{pmatrix}k\\-5\end{pmatrix}$
2(b)	Enlargement [SF] 3 [Centre] (1, 4)	**3**	**B1** for each
3(a)(i)	43	**1**	
3(a)(ii)	65	**1**	
3(a)(iii)	13	**1**	

Question	Answer	Marks	Partial Marks
3(b)	80	**3**	**M2** for $\frac{400}{18}\times\frac{60\times60}{1000}$ oe Or **M1** for $\frac{400}{18}$ or for *their* speed in m/s $\times\frac{60\times60}{1000}$ or for $\frac{400}{1000}$ and $\frac{18}{60\times60}$ soi
4(a)(i)	$\frac{1}{11}$ oe	**1**	
4(a)(ii)	$\frac{1}{110}$ oe	**2**	**M1** for $\frac{1}{11}\times\frac{1}{10}$ oe
4(a)(iii)	$\frac{4}{55}$ oe	**3**	**M2** for $\left(\frac{2}{11}\times\frac{1}{10}\right)+\left(\frac{3}{11}\times\frac{2}{10}\right)$ oe **or M1** for $\left(\frac{2}{11}\times\frac{1}{10}\right)$ or $\left(\frac{3}{11}\times\frac{2}{10}\right)$ seen oe
4(b)(i)	$\frac{1}{165}$ oe	**2**	**M1** for $\frac{3}{11}\times\frac{2}{10}\times\frac{1}{9}$ oe
4(b)(ii)	$\frac{1}{5}$ oe	**5**	**M4** for $3\left(\frac{2}{11}\times\frac{1}{10}\times\left[\frac{9}{9}\right]\right)+3\left(\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}\right)$ oe or **M3** for $3\left(\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}\right)$ or **M2** for $3\left(\frac{2}{11}\times\frac{1}{10}\times\left[\frac{9}{9}\right]\right)$ or $\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}$ oe or **M1** for $\frac{2}{11}\times\frac{1}{10}\times\left[\frac{k}{9}\right]$ where k is 3, 6 or 9
4(b)(iii)	$\frac{131}{165}$ oe	**2**	**M1** for 1 – (*their* **(b)(i)** + *their* **(b)(ii)**) oe
5(a)(i)	81° Angle at centre is twice angle at circumference oe	**2**	**B1** for 81°
5(a)(ii)	81° Alternate segment [theorem] oe	**2**	**FT** *their* **(a)(i)** **B1FT** for 81°